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HOUSE OF ASPIRANTS

COMPUTER

CHAPTER 7

Computer System

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CHAPTER 7

Computer System

Computer Awareness

IN THIS CHAPTER

- Hardware and software
- Motherboard components
- Power supply
- Storage accessories
- Computer ports
- Chipsets

- Intel Core i9

2.AMD

Examples:

- Ryzen 3
- Ryzen 5
- Ryzen 7
- Ryzen 9

MOST IMPORTANT FACTS FOR COMPETITIVE EXAMS

- CPU is called the Brain of the Computer.
- CPU stands for Central Processing Unit.
- Intel 4004 was the first microprocessor.
- Intel 4004 was introduced in 1971.
- Control Unit controls all activities.
- ALU performs arithmetic and logical operations.
- Registers are the fastest memory.
- Clock speed is measured in GHz.
- Program Counter stores the address of the next instruction.
- Instruction Cycle = Fetch → Decode → Execute → Store.
- Intel and AMD are major CPU manufacturers.

A computer system consists of two main components:

1.Hardware

2.Software

Hardware and software work together to perform tasks. Without hardware, software cannot run, and without software, hardware cannot function effectively.

HARDWARE

Hardware refers to all the physical and tangible components of a computer system that can be seen and touched.

Hardware is the physical part of a computer that we can see, touch, and use.

Examples

- Monitor
- Keyboard
- Mouse
- CPU
- Motherboard
- RAM
- ROM
- Hard Disk
- Printer
- Speakers

HARDWARE VS SOFTWARE

Hardware	Software
Physical components	Programs and instructions
Can be touched	Cannot be touched
Example: Keyboard	Example: MS Word
Subject to physical damage	Cannot be physically damaged

MOTHERBOARD

The Motherboard is the main circuit board of a computer that connects and holds together almost all internal hardware components.

It acts as the communication center of the computer.

Other Names

- Mainboard
- System Board
- Printed circuit board (PCB): the motherboard is a type of PCB; not every PCB is a motherboard.

Important Exam Point

Motherboard = Heart/Backbone of Computer System

Functions of Motherboard

1. Connects All Hardware

Connects:

- CPU
- RAM
- ROM
- Hard Disk
- Expansion Cards

2. Supplies Power

Distributes electrical power to all connected devices.

3. Enables Communication

Allows hardware components to communicate with one another.

4. Houses CPU

Provides a socket for installing the processor.

COMPONENTS OF A MOTHERBOARD

1. CPU Socket (CPU Slot)

A special socket provided on the motherboard for installing the CPU.

Exam Point

CPU is installed in the CPU Socket.

2. Memory Slots

Slots used for installing memory modules.

Used For

- RAM
- ROM Modules (in some systems)

Features

3. Expansion Slots

Expansion slots allow additional hardware cards to be installed.

Examples

- Graphics Card
- Sound Card
- Network Card
- TV Tuner Card

Exam Point

Expansion Slot = Used to add new hardware capabilities

4. Northbridge

Northbridge is a chipset responsible for communication between:

- CPU
- RAM
- Graphics Controller

Function

Handles high-speed communication inside the motherboard.

Note

Modern processors often integrate Northbridge functions into the CPU itself.

5. USB Port(Universal Serial Bus)

Uses

Connects devices such as:

- Keyboard
- Mouse
- Printer
- Pen Drive
- Scanner

6. PCI Slot(Peripheral Component Interconnect)

Connects:

- Sound Cards
- Modems
- Network Cards
- Video Cards

7. AGP Slot(Accelerated Graphics Port)

Specially designed for: Graphics Cards

Exam Point

AGP = Graphics Card Slot

8. Heat Sink

A metallic component that absorbs and dissipates heat generated by the processor.

Function

Prevents CPU overheating.

Often Used With

- Cooling Fan
- Thermal Paste

9. Power Connector

Supplies electrical power from the SMPS to the motherboard.

10. CMOS Battery(Complementary Metal Oxide Semiconductor)

The battery supplies power to the real-time clock and, in older designs, CMOS memory that holds settings such as:

- Date
- Time
- BIOS Settings
- Hardware Configuration

even when the computer is switched off. Exam Point

The CMOS battery supports the real-time clock and legacy CMOS settings; modern firmware settings may be stored in non-volatile flash.

11. SMPS(Switched Mode Power Supply)

SMPS converts AC power from the wall socket into DC power required by computer components.

Functions

- Controls power supply
- Protects components
- Supplies correct voltage

Exam Point

SMPS = Power Supply Unit of Computer

MICROPHONE

A Microphone is an input device that converts sound waves into electrical signals.

Emile Berliner developed an early microphone in 1877; he was one of several microphone pioneers.

Uses

- Voice Recording
- Video Conferencing
- Online Classes

- Audio Communication

USB FLASH DRIVE (PEN DRIVE)

A USB Flash Drive is a removable storage device that uses flash memory. Features

- Portable
- Rewritable
- USB Interface

Uses

- Data Transfer
- Backup
- Portable Storage

Exam Point

Pen Drive uses Flash Memory Technology.

MEMORY CARD

A Memory Card is a portable storage device used to store digital data.

Also Known As

- Flash Memory Card
- SD Card
- MicroSD Card

Uses

- Smartphones
- Cameras
- Camcorders
- Printers
- Gaming Consoles

Types

- SD Card
- MicroSD Card
- MiniSD Card

COMPUTER PORTS

Ports are connection points on a computer where external devices are connected.

Ports act like "gates" through which data enters or leaves the computer.

Types of Computer Ports

1. Hard Disk Ports

A. PATA(Parallel Advanced Technology Attachment)

Features

- Older technology
- Parallel data transfer

B. SATA(Serial Advanced Technology Attachment)

Features

- Faster than PATA
- Widely used today

Exam Point

SATA replaced PATA in modern computers.

C. SCSI(Small Computer System Interface)

Uses

- Servers
- High-performance storage systems

2. Parallel Port

Uses

- Printers
- Scanners

Features

Transfers multiple bits simultaneously.

3. Serial Port

Uses

- Modems
- Projectors
- Older Communication Devices

Features

Transfers data one bit at a time.

Parallel Port vs Serial Port

Parallel Port	Serial Port
Multiple bits at once	One bit at a time
Faster in old systems	Simpler design
Used for printers	Used for modems

4. PS/2 Port

Uses

- Keyboard
- Mouse

Developed By IBM

Exam Point

PS/2 Port = Mouse and Keyboard Connection

5. VGA Port(Video Graphics Array)

Uses

Connects:

- Monitor
- Projector

Signal Type

Analog Video Signal

6. HDMI Port(High Definition Multimedia Interface) Uses

Transfers:

- Audio
- Video

through a single cable.

Devices

- TV
- Monitor
- Projector
- Gaming Console

Exam Point

HDMI carries both Audio and Video.

7. FireWire Port

Developed By Apple

Uses

- High-Speed Data Transfer
- Digital Cameras
- External Storage Devices

Other Name

IEEE 1394

8. Ethernet Port

Used for connecting computers to a network Also Called LAN Port Uses

- Internet Connection
- Local Area Network (LAN)

Exam Point

Ethernet Port = LAN Port

9. AGP Port(Accelerated Graphics Port)

Purpose

Dedicated graphics card connection.

IMPORTANT PORTS AT A GLANCE

Port	Full Form	Use
USB	Universal Serial Bus	Mouse, Keyboard, Pen Drive
HDMI	High Definition Multimedia Interface	Audio & Video
VGA	Video Graphics Array	Monitor
PS/2	Personal System/2	Keyboard, Mouse
Ethernet	Network technology name; LAN port is a descriptive name, not its full form.	Internet
SATA	Serial Advanced Technology Attachment	Hard Disk
AGP	Accelerated Graphics Port	Graphics Card
FireWire	IEEE 1394	High-Speed Data Transfer

Chipset

A chipset controls communication between:

- CPU
- RAM
- Storage Devices
- Expansion Cards

MOST IMPORTANT FACTS FOR COMPETITIVE EXAMS

- Hardware refers to physical computer components.
- Motherboard is called the Heart or Backbone of the Computer.
- The motherboard is a printed circuit board (PCB).
- Intel CPU is installed in CPU Socket.
- The CMOS battery powers the real-time clock and legacy settings memory.
- SMPS controls power supply.
- USB = Universal Serial Bus.
- HDMI carries audio and video together.
- VGA is used for monitors.
- Ethernet Port is also called LAN Port.
- SATA replaced PATA.
- FireWire is also called IEEE 1394.
- AGP is used for graphics cards.

ONE-LINE REVISION POINTS

- Hardware means physical computer parts.
- Motherboard connects all hardware.
- CPU is installed in CPU socket.
- Expansion slots add new hardware.
- CMOS stores BIOS settings.
- SMPS supplies power.
- USB is the most common port.
- HDMI transfers audio and video.
- Ethernet is used for networking.
- SATA is used for hard drives.
- AGP is used for graphics cards.

QUICK REVISION NOTES (30 SECONDS BEFORE EXAM)

- Hardware = Physical Components
- Motherboard = Heart of Computer
- Main Components:

- CPU Socket

- Memory Slots

- Expansion Slots

- CMOS Battery

- SMPS

- Ports:

-→ USB Mouse, Keyboard

-→ HDMI Audio + Video

-→ VGA Monitor

- PS/2 → Keyboard & Mouse

- Ethernet → LAN

- SATA → Hard Disk

-→ AGP Graphics Card

-→ CMOS Stores BIOS Settings

-→ SMPS Power Supply Unit

SOFTWARE